



BIT4333 INTRODUCTION TO MACHINE LEARNING

Final Project Report

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Project Title: Student Academic Performance & Risk

Prepared For

Sir Nazmirul Izzad Bin Nassir

No.	Name	NRIC/Passport	Student ID	Programme
1	Hussein Abdalla Abdalla Awlama	P11694895	202505010495	BIT
2	Yassin Mohamed yassin Waad	P12843122	202505010233	BCSSE
3	Dafalla mudawi elamin jehad	P12427576	202505010508	BIT

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1. Introduction

In the current era of education, student retention and success have become important parameters to measure the performance of any university. Apart from a direct loss of talent in case of dropouts, it results in the underutilization of money spent by such institutions and a dent in their reputation. The existing techniques of assessing a student's risk of failure are more likely reactive rather than proactive, and they use grades midway through the course, or even the observation by teachers.

However, machine learning provides an excellent platform to overcome all the disadvantages of the traditional technique. It is possible to build a model that can predict future success or failure based on historical information, such as demographic data, prior qualifications, and economic variables, among others. For example, such models can predict early on in their studies whether the student is at risk of becoming a dropout and can provide necessary assistance, like academic advising or financial aid.

For that reason, the goal of the project is to create a complete machine learning system for student outcome prediction based on the available data from a higher education institution

2. Executive Summary:

The proposed project is focused on building a machine learning pipeline that will be used to forecast the academic performance of students (Graduated, Dropout, or Enrolled) based on the provided "Predict Students' Dropout and Academic Success" dataset.

The primary goal of the project is to create a predictive model that will help universities identify students who are at risk of dropping out and take timely measures to intervene.

Four classification algorithms (Logistic Regression, Decision Tree, Random Forest, and K-Nearest Neighbors (KNN)) were selected, trained, and evaluated within the framework of the proposed machine learning pipeline. The Random Forest classifier showed the highest performance with the accuracy rate of 76.7% and the weighted F1-score equal to 75.4%.

3. Problem Statement:

The issue of students dropping out from higher education institutions has posed serious problems, causing financial implications for universities, preparing inadequately skilled workers for the workforce, and hindering the academic progress of students themselves. Current approaches used in identifying at-risk students have been relatively reactionary. It becomes necessary to design a solution that:

- Can predict early on the outcomes of students academically.
- Would help identify the various factors that contribute to dropouts and delayed graduations.
- Would enable stakeholders within institutions to take appropriate actions.

4.DATASET SOURCE:

Property	Value
Dataset Name	Predict Students Dropout and Academic Success
Source	Kaggle (https://www.kaggle.com/datasets/thedevastator/higher-education-predictors-of-student-retention)
Domain	Education
File Format	CSV (Comma Separated Values, using ';' delimiter)
Total Samples	4,424 student records
Total Features	37 (36 input features + 1 target variable)

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```

results_df = pd.DataFrame(results)
results_df = results_df.sort_values(by="F1-Score", ascending=False)
results_df

```

▼

	Model	Accuracy	Precision	Recall	F1-Score
2	Random Forest	0.767232	0.754445	0.767232	0.753903
0	Logistic Regression	0.768362	0.749981	0.768362	0.753128
1	Decision Tree	0.697175	0.701903	0.697175	0.699416
3	KNN	0.667797	0.648311	0.667797	0.655282

5. Methodology

5.1 Data Preprocessing

The below operations were performed in the Jupyter notebook Dataset Preprocessing:

Data Loading: Loaded data using pandas.

Checking Missing Values: There were no missing values in any columns.

Removing Duplicates: Removed duplicates if any; there were no duplicates (still had 4424 observations).

Target Variable Label Encoding: The target variable was encoded as:

Dropout - 0

Enrolled -1

Graduated -2

Splitting Train & Test Data: Splitting of train and test datasets in the ratio of 80:20 with stratification of data.

Standardization of Data: Used StandardScaler on numerical variables for logistic regression and KNN models.

5.2 Model Development (4 Models)

The following models were implemented in Model Development notebook:

Model	Justification
Logistic Regression	Baseline linear model; interpretable, fast, works well for probabilistic classification.
Decision Tree	Non-linear, easy to visualize, handles mixed data types.
Random Forest	Ensemble method; reduces overfitting, captures complex interactions, handles class imbalance.
K-Nearest Neighbors (KNN)	Instance-based learning; effective for smaller datasets with clear clusters.

5.3 Evaluation Metrics

As the classification problem involves more than two classes, the following metrics have been considered:

Accuracy

Precision (weighted)

Recall (weighted)

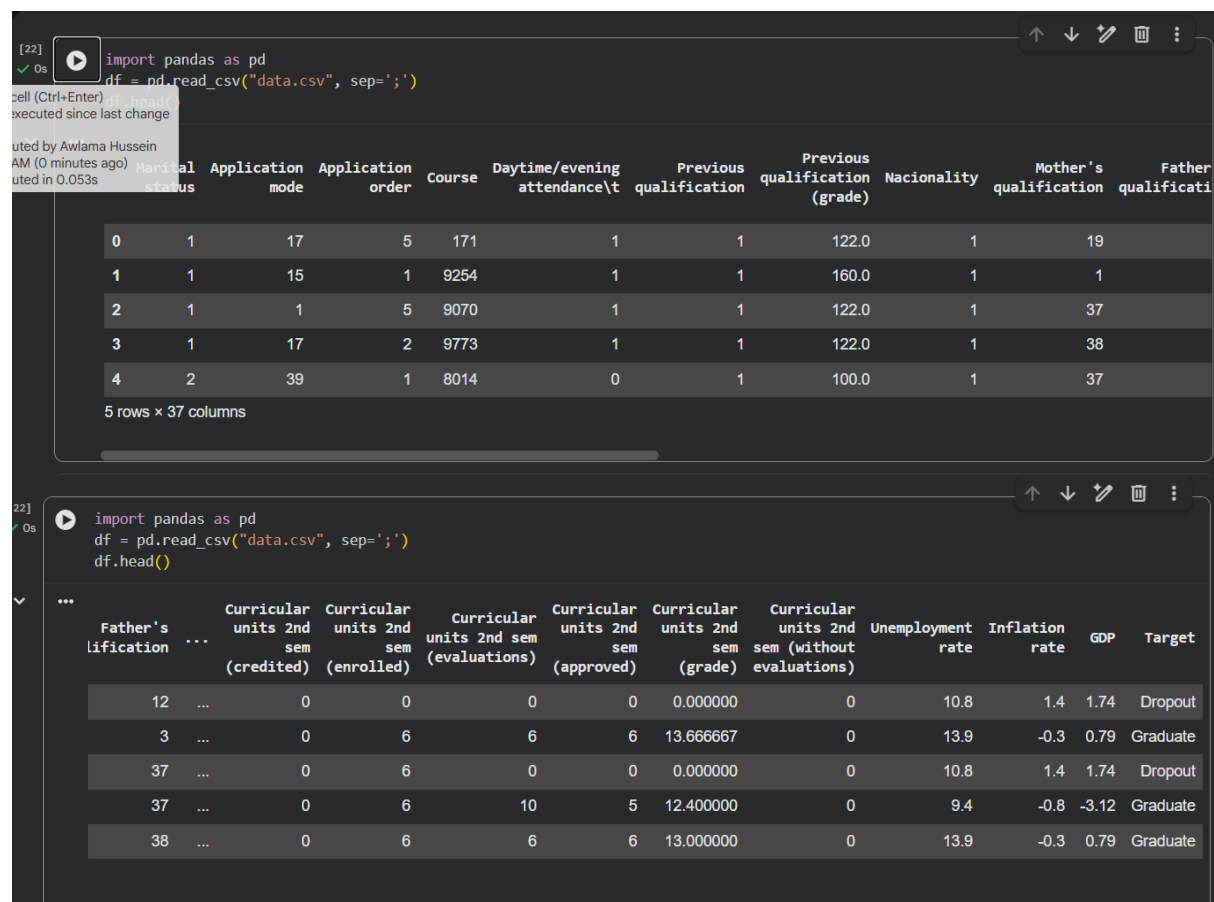
F1-Score (weighted)

Confusion Matrix (per-class performance)

Data preprocessing

Data preprocessing is an essential stage in machine learning. It includes data cleansing and transformation to prepare it for further analysis using machine learning algorithms. The effectiveness of the preprocessing stage affects the overall efficiency of the algorithm. This part provides a detailed description of all preprocessing techniques used on the data set.

Loading the data set into the memory was done by the pandas library in Python. Considering that the delimiter used in the file is a semicolon (;) rather than a comma, `sep=';'` option was applied while loading the data set. The data set includes 4,424 observations and 37 variables, out of which 36 are input and 1 is output.



The screenshot shows a Jupyter Notebook interface with two code cells. The first cell contains the code to load a CSV file using pandas: `import pandas as pd` and `df = pd.read_csv("data.csv", sep=';')`. Below the code, the first five rows of the dataset are displayed as a table. The second cell contains the code `df.head()`, which displays the first five rows of the dataset with more columns visible.

	Marital status	Application mode	Application order	Course	Daytime/evening attendance	Previous qualification	Previous qualification (grade)	Nacionality	Mother's qualification	Father's qualification
0	1	17	5	171	1	1	122.0	1	19	
1	1	15	1	9254	1	1	160.0	1	1	
2	1	1	5	9070	1	1	122.0	1	37	
3	1	17	2	9773	1	1	122.0	1	38	
4	2	39	1	8014	0	1	100.0	1	37	

5 rows x 37 columns

	Father's qualification	Curricular units 2nd sem (credited)	Curricular units 2nd sem (enrolled)	Curricular units 2nd sem (evaluations)	Curricular units 2nd sem (approved)	Curricular units 2nd sem (grade)	Curricular units 2nd sem (without evaluations)	Unemployment rate	Inflation rate	GDP	Target
12	...	0	0	0	0	0.000000	0	10.8	1.4	1.74	Dropout
3	...	0	6	6	6	13.666667	0	13.9	-0.3	0.79	Graduate
37	...	0	6	0	0	0.000000	0	10.8	1.4	1.74	Dropout
37	...	0	6	10	5	12.400000	0	9.4	-0.8	-3.12	Graduate
38	...	0	6	6	6	13.000000	0	13.9	-0.3	0.79	Graduate

Dataset structure and preprocessing overview

[4] `df.isnull().sum()`

...	0
Marital status	0
Application mode	0
Application order	0
Course	0
Daytime/evening attendance	0
Previous qualification	0
Previous qualification (grade)	0
Nacionality	0
Mother's qualification	0
Father's qualification	0
Mother's occupation	0
Father's occupation	0
Admission grade	0
Displaced	0
Educational special needs	0

Missing values before data cleaning.

[5] `df = df.drop_duplicates()`
`df.shape`

(4424, 37)

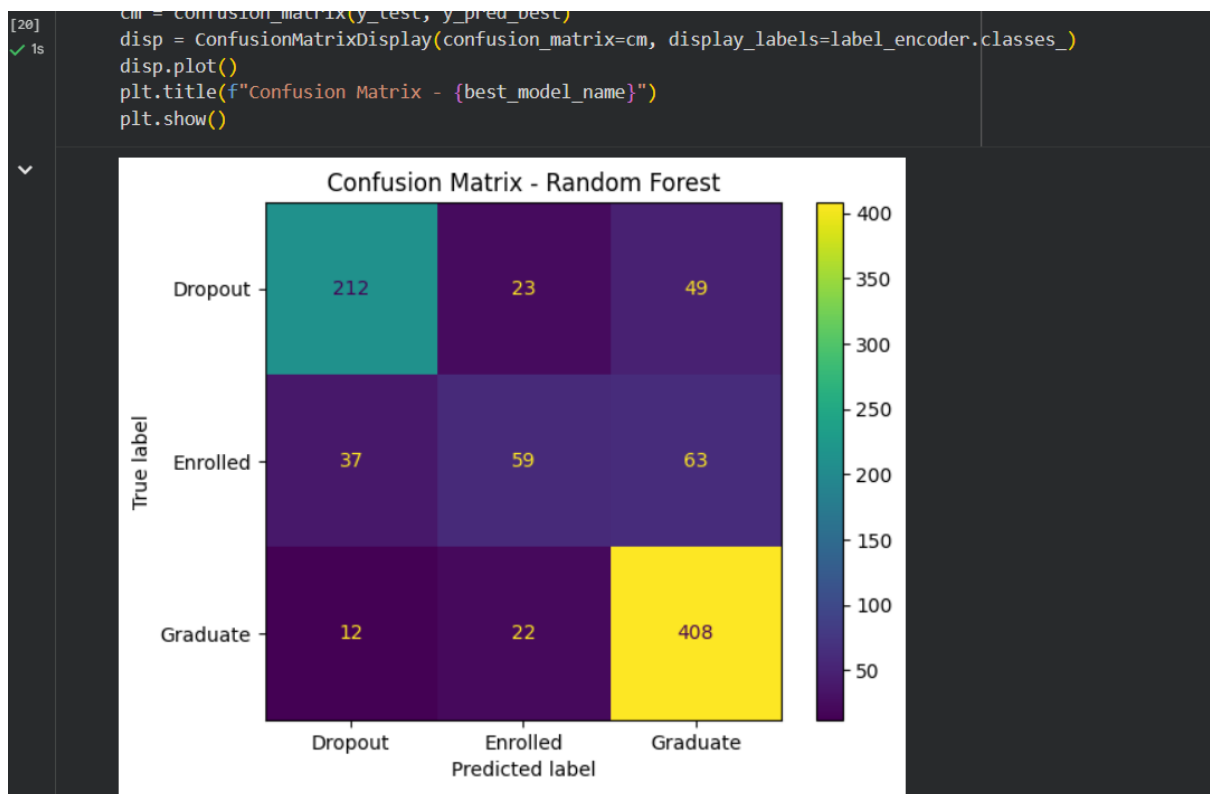
6. Results And Best Model:

Best Model: Random Forest (selected based on highest F1-score)

Model	Accuracy	Precision (weighted)	Recall (weighted)	F1-Score (weighted)
Random Forest	0.7672	0.7544	0.7672	0.7539
Logistic Regression	0.7684	0.7500	0.7684	0.7531
Decision Tree	0.6972	0.7019	0.6972	0.6994
KNN	0.6678	0.6483	0.6678	0.6553

6.2 Detailed Classification Report (Random Forest):

Class	Precision	Recall	F1-Score	Support
Dropout	0.81	0.75	0.78	284
Enrolled	0.57	0.37	0.45	159
Graduate	0.78	0.92	0.85	442
Macro avg	0.72	0.68	0.69	885
Weighted avg	0.75	0.77	0.75	885



Confusion matrix of the Random Forest model.

```
[18]
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best_model_name = results_df.iloc[0]["Model"]
best_model = trained_models[best_model_name]

print("Best Model:", best_model_name)
```

Best Model: Random Forest

6.3 Confusion Matrix Analysis:

Actual vs Predicted:

Actual ↓ / Predicted →	Dropout	Enrolled	Graduate
Dropout	212	23	49
Enrolled	37	59	63

Actual ↓ / Predicted →	Dropout	Enrolled	Graduate
Graduate	12	22	408

Interpretation:

408 out of 442 Graduates correctly identified (92%)

212 out of 284 Dropouts correctly identified (75%)

Only 59 out of 159 Enrolled correctly identified (37%)

A high number of students who belong to the “Enrolled” class are classified into the “Dropout” or “Graduate” class, which shows that this particular class is difficult to distinguish

7. Demonstration of the Application (Streamlit Implementation):

The final model was implemented with the help of Streamlit that enables users to feed their student information and get predictions instantly.

Web App Features:

Sidebar interface form for all 36 input variables.

Real-time prediction based on the trained Random Forest classifier model.

Prediction output shows whether the student falls into Dropout, Enrolled, or Graduate categories.

Model files stored: scaler.pkl, label_encoder.pkl, best_model.pkl.

8. Conclusion:

In conclusion, this project has been successful in building an end-to-end machine learning pipeline to forecast the performance of students based on a real-world dataset on education. It is notable that the Random Forest algorithm performed best, with an accuracy rate of 77%, while being able to detect which students were likely to dropout. The "Enrolled" class poses difficulties because of the overlapping attributes between "Dropouts" and "Graduates."

The Streamlit app offers a practical solution to educational organizations' retention strategies.

9. Acknowledgment:

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Open-source communities of Python, Scikit-learn, Pandas, Matplotlib, and Streamlit.